AIA Healthcare Insurance Policy for VPINS

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Problem understanding

VPINS is a division in VPB, VPINS is a division in VPB, aims to introduce a specialized healthcare insurance policy called AIA policy. This initiative seeks to offer a comprehensive insurance plan tailored to the diverse needs of its customer base, focusing on affordability, extensive coverage, and ease of access.

Tasks:

- Predict customers likely to buy the AIA policy (output: top 800 IDs).
- Provide actionable insights into customer behaviour and decision-making.

1. Business Requirement

1. **Expand Insurance Offerings:** Enhance VPBank's product portfolio with a unique healthcare insurance plan.
2. **Customer Retention:** Provide value-added services to improve customer loyalty.
3. **Revenue Growth:** Generate a new revenue stream through insurance premiums and partnership incentives.

2. Data Processing

1 **Training dataset**
(train_data.txt)

86 attributes:
- Sociodemographic (1-42)
- Product ownership (43-85)
- Target label: Policy possession (86)

2 **Testing dataset**
(test_data.txt)

Same format as training but without labels.

3 **Attribution Decription**

Detailed explanations provided in supplementary file.

Approach

01 Data Understanding

- Analyzed sociodemographic and product features.
- Explored class balance and correlations.

02 Data Preparation

- Imputed missing values.
- Normalized and encoded categorical variables.

03 Feature Engineering

- Derived meaningful features.
- Selected top attributes using feature importance scores.

3. Data exploration and Modelling workflow

1. **Baseline Models:**

   Logistic Regression and Decision Trees for initial evaluation.

2. **Advanced Models:**

   Gradient Boosting Machines (XGBoost, LightGBM).

3. **Hyperparameter Tuning**: Optimized using grid and random search.

4. **Validation**: cross-validation to assess generalizability.

Result

Model Performance:

- Best Model: XGBoost
    - Accuracy: 0.93.
    - Precision: 0.2
    - Recall: 0.049
    - AUC-ROC: 0.93

Predictions: Generated top 800 promising customers for AIA policy with the file attached.

Insights & Interpretability

Customer Insights:

- Key sociodemographic drivers: Average income (feature 42), National Health Service (feature 35) and Customer Subtype (feature 1)
- Product ownership trends: High likelihood for customers with complementary insurance products.

Actionable Insights:

- Target focus on Car policies (feature 47) and Fire policies (feature 59)
- Design cross-selling campaigns leveraging existing product ownership.

Conclusion

1. Challenges:

- Imbalanced dataset.
- Limited interpretability of complex models.

2. Future Improvements:

- Incorporate external data (e.g., economic trends).
- Deploy explainable AI techniques for enhanced transparency.

3. Deliverables:

- CSV file with 800 customer IDs.
- Codebase for reproducibility (Python scripts).
- Documented insights for marketing strategy.